

6. Project Testing Phase

1. INTRODUCTION

The Project Testing Phase validates the end-to-end functionality of the **ComicCraft** application, verifying the complete workflow from initial user input through to generated comic pages and final PDF export.

2. OBJECTIVES

Phase 6 focuses on comprehensive quality assurance and endpoint verification. Key objectives include:

- Test web application entry points and homepage delivery.
- Validate input forms and enforce required field logic.
- Exercise the primary /generate route across diverse payloads.
- Test underlying JSON REST API responses.
- Verify image generation and prompt pipeline execution.
- Verify automated layout synthesis and page structure assembly.
- Validate PDF export formatting and output file integrity.
- Verify edge-case handling for invalid inputs and server errors.
- Confirm deployment and endpoint accessibility on Uvicorn.

3. PHASE WORKFLOW

Testing runs sample workflows across positive and negative paths: valid inputs proceed completely through the generation pipeline, while missing or malformed attributes trigger immediate validation logic or HTTP error responses.

4. DETAILED EXPLANATION

Execution testing utilizes Uvicorn for local server orchestration and interactive validation via OpenAPI specs at /docs. Automated test cases assert system reliability under varying input conditions, with pass/fail records preserved within repository test reports to verify student implementation status.

5. GITHUB-READY CODE / STRUCTURE

The following pytest suite verifies core application endpoints and validation behaviors:

```
from fastapi.testclient import TestClient
from app.main import app
```

```
client = TestClient(app)

def test_homepage():
    response = client.get("/")
    assert response.status_code == 200

def test_missing_prompt():
    response = client.post("/generate", data={
        "story_prompt": "",
        "character_name": "Alex",
        "setting": "Forest",
        "tone": "Funny",
        "art_style": "Comic Book"
    })
    assert response.status_code in (200, 400, 422)
```

6. GITHUB FILES & DELIVERABLES

Key Testing Artifacts:

- tests/ — Automated unit and integration test suites.
- docs/test-report.md — Detailed execution logs, pass/fail matrices, and coverage reports.
- **API Test Screenshots** — Verified endpoint execution captures from Swagger UI / Postman.

7. EXPECTED OUTCOME

At the conclusion of Phase 6, all primary routes, AI integrations, layout engines, and PDF generation tools within **ComicCraft** are systematically tested and documented. Actual completion verification relies on the execution results and repository status.

8. PHASE CONCLUSION

Phase 6 provides a structured foundation for the continuous deployment of **ComicCraft**, ensuring stability, robust error mitigation, and high-quality deliverables across GitHub-based project evaluation.